<u>Element K</u>

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K1 is summarizing each element

K2 is a value judgement, basically asking what went well and what went bad

K3 is general lessons learned related to engineering design

(This is just for organization)

Defining The Problem	**Selection & Prototype Design**	**Prototype Building**	**Testing**	**Evaluation**
A	D	G	H	J
B	E		I	
C	F			

Defining The Problem:

- Element A
  - Element A focuses on clearly identifying, explaining, and justifying Dr. Brossman's need for a device to measure the Maximum Inspiratory Pressure for her students. It emphasizes providing a well-researched, detailed explanation of why the problem exists, who it affects, and why it is important. This element also evaluates the depth of research and credibility of sources used to support the problem statement.

- Element B
    - In Element B, each group member researched different types of MIP respiratory devices. We assign each member to find the most affordable solution. Another member would see a device that would be the most accurate. After researching, we would list the pros and cons of each example we could use.
- Element C
    - Element C evaluates how well design requirements are defined, justified, and prioritized to create a viable solution. It assesses the relevance of requirements based on our research and Dr. Brossman's needs and prioritization to the final design. High-scoring responses demonstrate specific and objective criteria that effectively guide the solution.

Selection & Prototype Design:

- Element D
    - In this project phase, our group brainstormed what mechanics could cover what function. For displaying pressure, we suggested a digital display, a dial, or a barometer. Secondly, we came to a consensus for our decision matrix. Adjust restriction, sealed air, and restricted airflow are worth 30% when we rank each conceptual design. Display pressure was worth 10% for our grading rubric. Each member of our group then drew and sketched out a conceptual design.

- Element E
  - Element E evaluates how well STEM principles and expert input support the proposed solution. It assesses the application of relevant STEM concepts and the extent of expert verification.
- Element F
  - This element verified each of our priorities. Our group proved the device was reusable, and an image ensured no hazardous parts. The rest of the document covers each of the following priorities.

Prototype Building:

- Element G
  - Element G evaluates each construction process step, from buying the PVC pipe to our final product. Each step includes an image and a description of why and how we performed it.

Testing:

- Element H
  - Element H has two parts: our priorities and how to use the device. For each priority, we listed the step(s) the user must take to achieve it. The data for each priority follows. For repeatability, for example, the data would be the rise of the water in the PVC pipe. The second part of this document lists the parts, how to assemble the device if it isn't already, and how to use it safely.
- Element I
  - Similar to the previous elements, Element I covers the importance of tests. Each priority is ranked chronologically by importance, followed by a brief description.

The comprehension of the testing procedure follows this. Our group then recapped the experiments that worked and the ones that did not. For example, our attempt to calculate the negative pressure with a syringe failed.

Element K2

What went well?

The device went smoothly, and our main PVC pipe was easy to find. The shape was very reliable and helped us view what our concept design would look like and if it would work. The shape also helped 3d print the parts we needed and could not find in stores, which allowed us to come up with our final design. The research was easy to find on cleaning procedures, learning how a MIP device works, and finding the most affordable solution. Researching gave our group a good understanding of what was needed for our project. Another crucial part of our project that went well was the finding of our one-way valve part. The first one-way valve we had was very expensive, but we found another one-way valve that came in a larger quantity and was cost-effective.

What did not go well?

Initially, the project was supposed to be shaped as a "U." We would have used a PVC pipe the shape of a "U," inhaling on one side of the tube and having the water go down on the other PVC pipe. Our group then settled on using one tube with a wider diameter than two PVC pipes with smaller PVC pipes. If our group went with the original design, our device would be very tall because the water has an easier time traveling through a smaller space.

Because of the nature of our final project, calibrating and finding a discrete number to signal the strength of one's inhale was not achieved. However, having the water rise was sufficient enough to show how powerful the inhale is. The stronger the inhale, the higher the water will rise.

Element K3

What lessons were learned related to engineering design?

This project educated us about the main components of the project: what a MIP device is, and how to make one. As our group embarked on the project, we learned and equipped a lot of skills related to engineering design.

Our project management has greatly improved along with teamwork and collaboration. At the beginning of the project our organization was all over the place, we still had a lot to figure out and put together. As the project progressed, our planning was on point regarding what we wanted to accomplish each week. It made it easier for us when each person of the group had a specialty so whenever we needed something we could get it done. Not only have we grown as a group but we have also made sure to keep each other accountable to make sure we are on top of things, which also shows our determination to maintain time management.

Our creativity was great for this project because it allowed us to create something that works and is also what we wanted to create. Creating projects with requirements and a goal in mind allows us to use our creativity in the best way possible. The elements have allowed us to precisely make and meet our requirements to the best of our abilities.